

Internship Program Name: Cyber Security & Ethical Hacking

Task No.: 03

Task Name: Vulnerability Scanning & Assessment

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Vulnerability Scanning & Assessment

Aim: To perform vulnerability scanning and assessment on a controlled lab environment using Kali Linux and cybersecurity tools in order to identify open ports, web vulnerabilities, and security misconfigurations safely and ethically.

Objective:

The purpose of this task is to help you understand how cybersecurity analysts and penetration testers identify weaknesses in a system before attackers do.

Objective	Real-world outcome
Perform network reconnaissance	Understand how attackers discover targets
Identify open ports & services	Detect exposed attack surfaces
Detect misconfigurations	Learn what makes systems vulnerable
Use scanners responsibly	Learn professional assessment workflow
Prepare a VA Report	Document findings like a Cybersecurity Analyst

1. Introduction:

This report presents the results of a vulnerability assessment conducted in a controlled and isolated cybersecurity lab environment. The objective of this task was to identify potential weaknesses in a system before attackers could exploit them.

The assessment was performed on a locally hosted vulnerable web application (DVWA) using Kali Linux within Oracle VirtualBox. Various tools such as Nmap, Nikto, and Dirsearch were used to identify open ports, services, web vulnerabilities, and hidden directories.

The findings revealed that the web server is active and accessible, with certain security misconfigurations such as missing security headers and exposed services. No exploitation was performed, as the task focused only on identification and documentation.

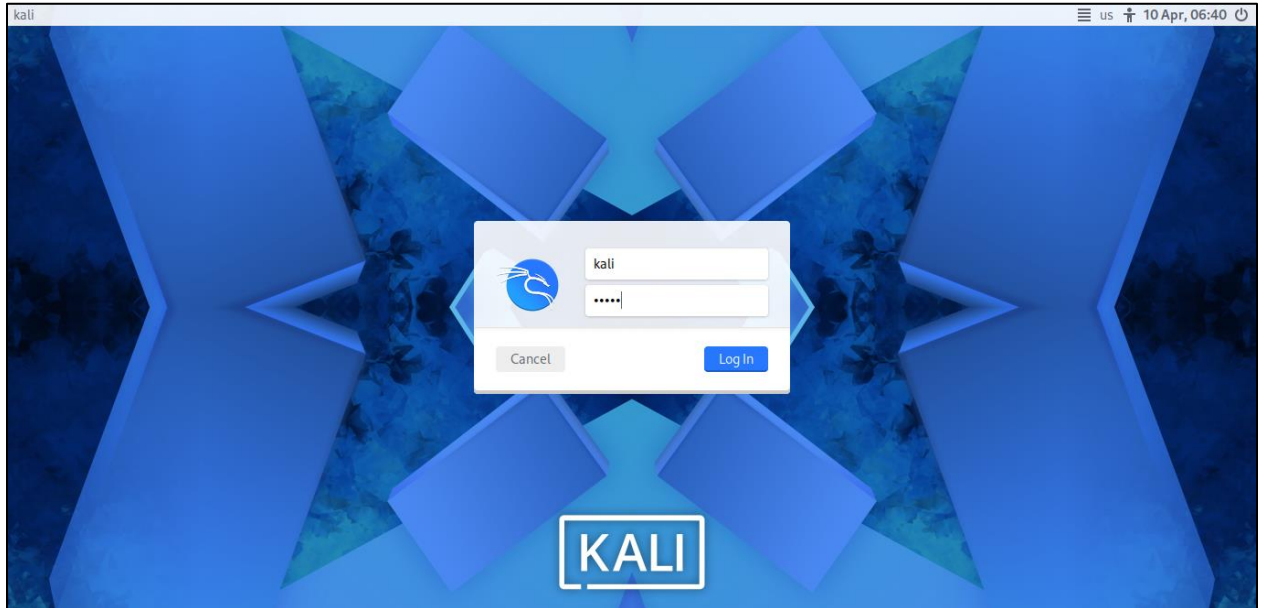
2. Lab Environment Setup

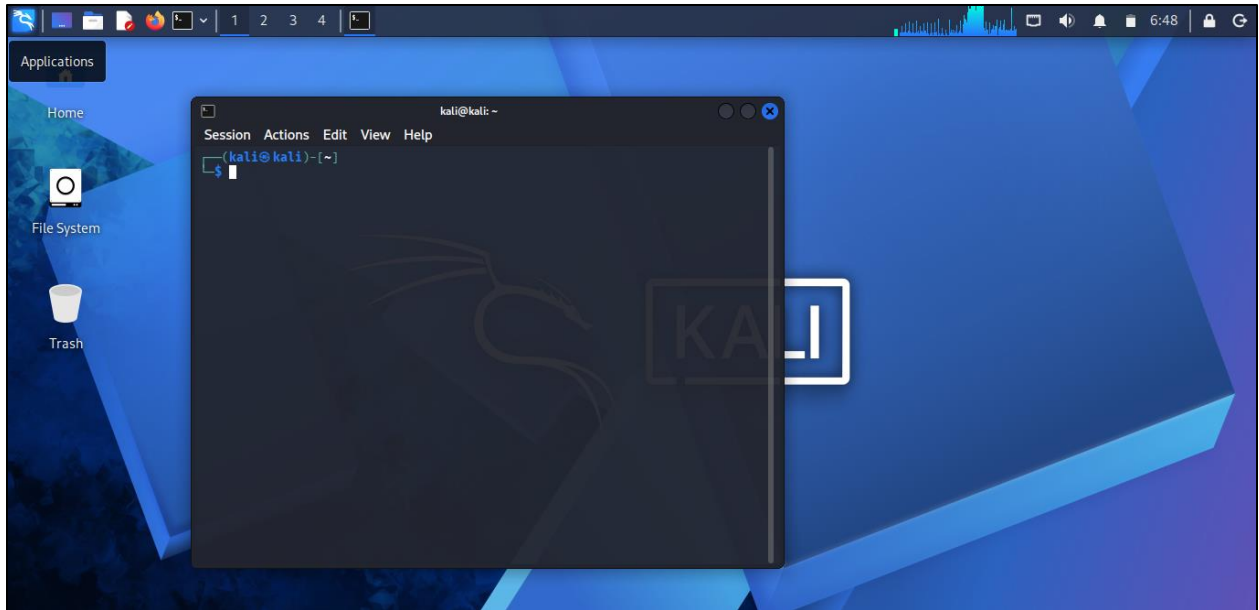
- **Host System:** Windows Laptop
- **Virtualization Tool:** Oracle VirtualBox
- **Virtual Machine:** Kali Linux
- **Target Application:** DVWA (Damn Vulnerable Web Application)
- **Environment Type:** Local isolated lab

3. Methodology (Step-by-Step Execution)

Step 1: Launching the Lab Environment

- Oracle VirtualBox was opened
- Kali Linux Virtual Machine was started
- User logged into Kali Linux

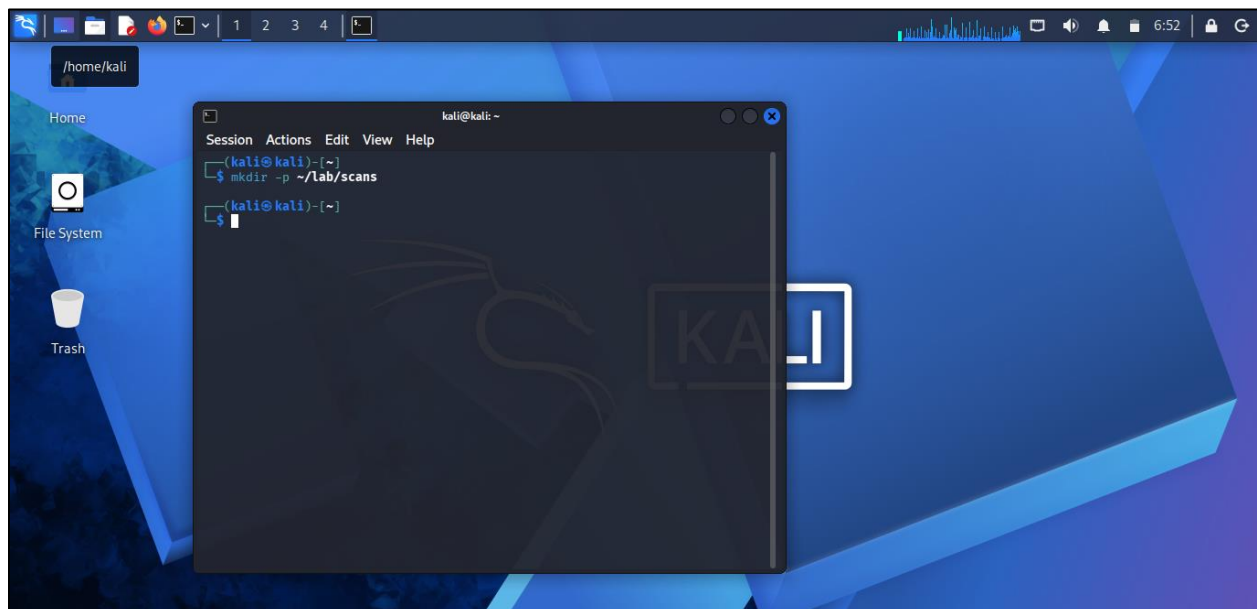




Step 2: Creating Directory for Scan Results

A folder was created to store all scan outputs:

```
mkdir -p ~/lab/scans
```



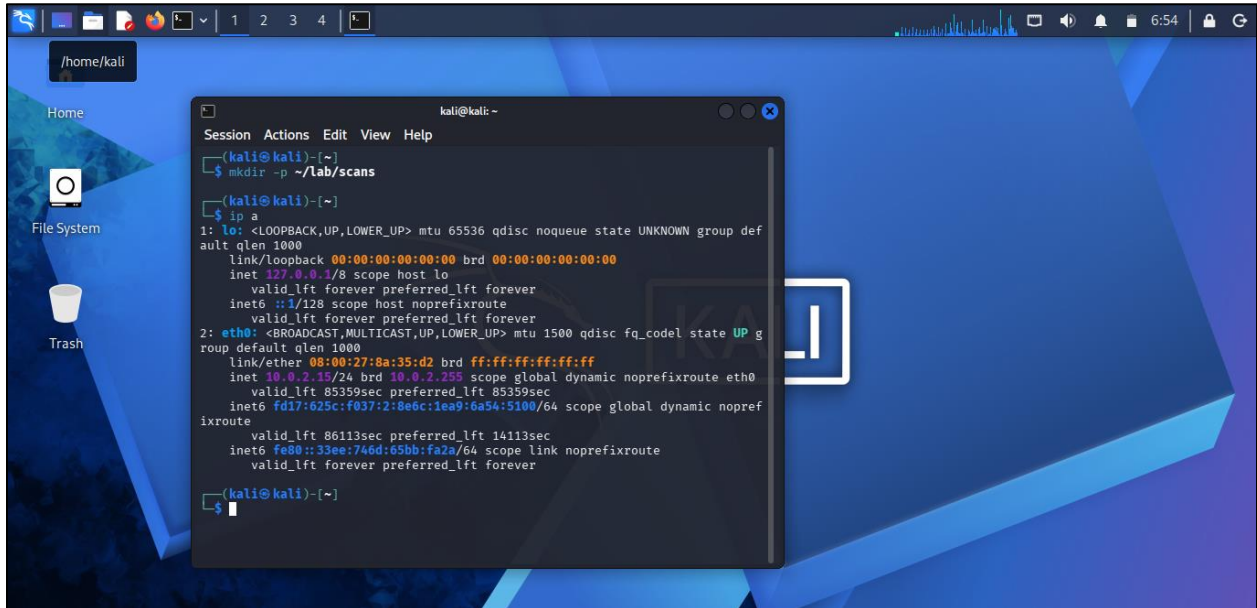
Step 3: Checking IP Configuration

The IP address of the system was verified:

```
ip a
```

Observation:

- Loopback IP: 127.0.0.1
- Network Interface: eth0



```
kali@kali: ~  
$ mkdir -p ~/lab/scans  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def  
ault qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
roup default qlen 1000  
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 85359sec preferred_lft 85359sec  
    inet6 fd17:625c:f037:2:8e6c:1ea9:6a54:5100/64 scope global dynamic nopref  
ixroute  
        valid_lft 86113sec preferred_lft 14113sec  
    inet6 fe80::33ee:746d:65bb:fa2a/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
$
```

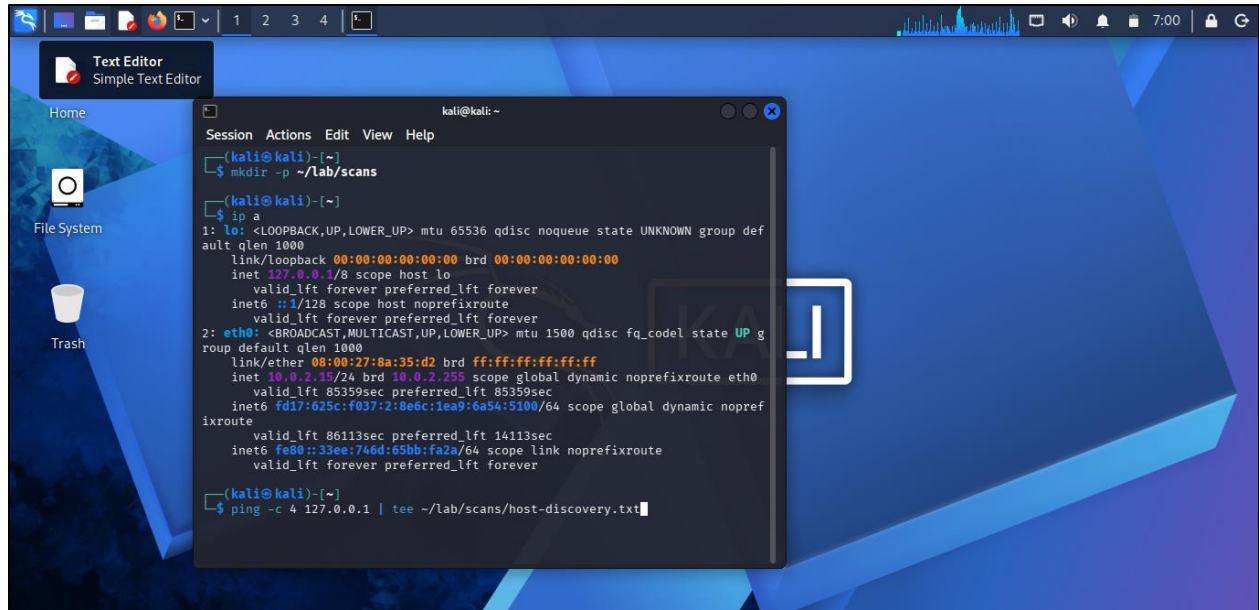
Step 4: Host Discovery (Connectivity Check)

The target system was tested for reachability:

```
ping -c 4 127.0.0.1 | tee ~/lab/scans/host-discovery.txt
```

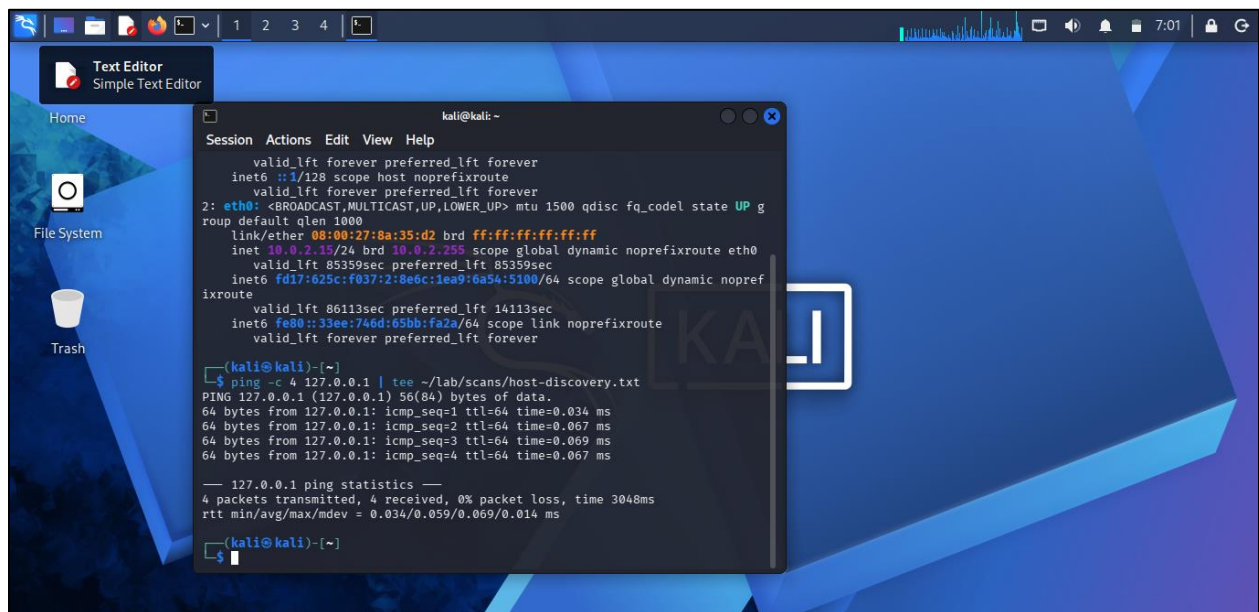
Observation:

- Target responded successfully
- 0% packet loss



The screenshot shows a Kali Linux desktop with a terminal window open. The terminal displays the following commands and output:

```
kali@kali:~$ mkdir -p ~/lab/scans
kali@kali:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
    ault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
    roup default qlen 1000
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 85359sec preferred_lft 85359sec
    inet6 fd17:625c:f037:2:8e6c:1ea9:6a54:5100/64 scope global dynamic nopref
        ixroute
        valid_lft 86113sec preferred_lft 14113sec
    inet6 fe80::33ee:746d:65bb:fa2a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
kali@kali:~$ ping -c 4 127.0.0.1 | tee ~/lab/scans/host-discovery.txt
```



The screenshot shows the same Kali Linux desktop environment. The terminal window now displays the output of the ping command:

```
kali@kali:~$ ping -c 4 127.0.0.1 | tee ~/lab/scans/host-discovery.txt
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data:
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.034 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.067 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.069 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.067 ms

--- 127.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3048ms
rtt min/avg/max/mdev = 0.034/0.059/0.069/0.014 ms
kali@kali:~$
```

Step 5: Port and Service Enumeration (Initial Attempt)

Initial Nmap scan was performed:

```
nmap -sV -p- 127.0.0.1 -oN ~/lab/scans/full-service-scan.txt
```



```
kali@kali:~$ ping 127.0.0.1
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.067 ms

--- 127.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3048ms
rtt min/avg/max/mdev = 0.034/0.059/0.069/0.014 ms

kali@kali:~$ nmap -sV -p- 127.0.0.1 -oN -lab/scans/full-service-scan.txt
Failed to open normal output file -lab/scans/full-service-scan.txt for writing: No such file or directory (2)

kali@kali:~$ mkdir -p /home/kali/lab/scans

kali@kali:~$ nmap -sV -p- 127.0.0.1 -oN /home/kali/lab/scans/full-service-scan.txt
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-10 07:18 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000025s latency).
All 65535 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 65535 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 5.30 seconds

kali@kali:~$
```

Step 6: Starting Apache Web Server

Apache service was started:

```
sudo service apache2 start
```

```
kali@kali:~$ sudo service apache2 start
[sudo] password for kali:
```

Service status was verified:

```
sudo service apache2 status
```

Observation:

- Apache server was active and running

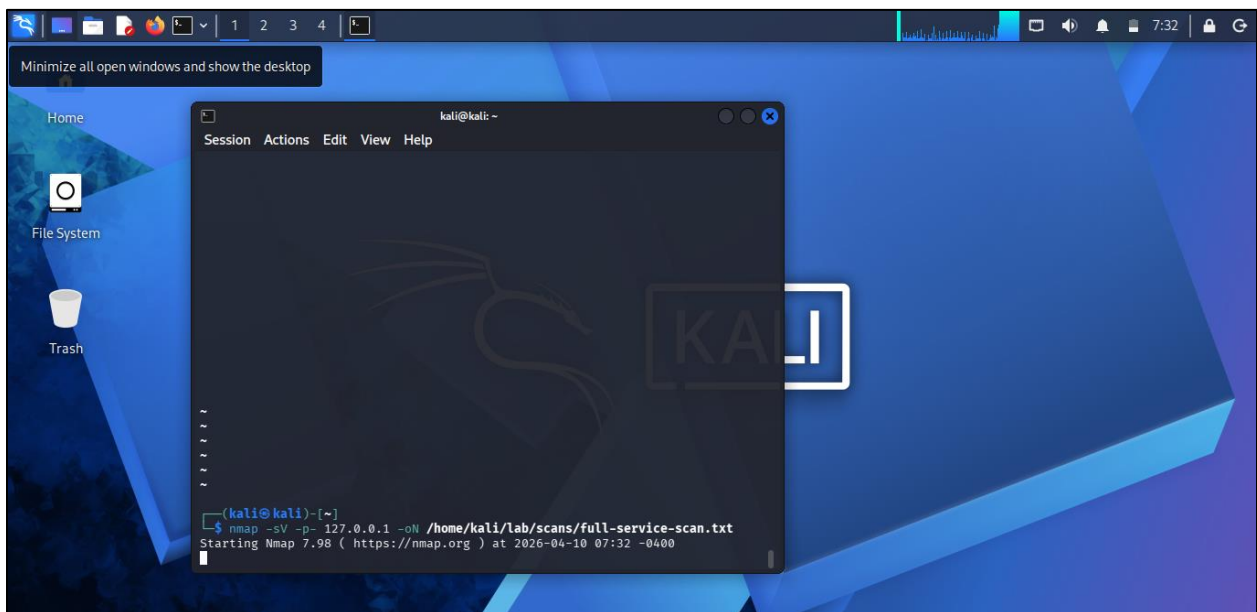
Step 7: Port and Service Enumeration (Final Scan)

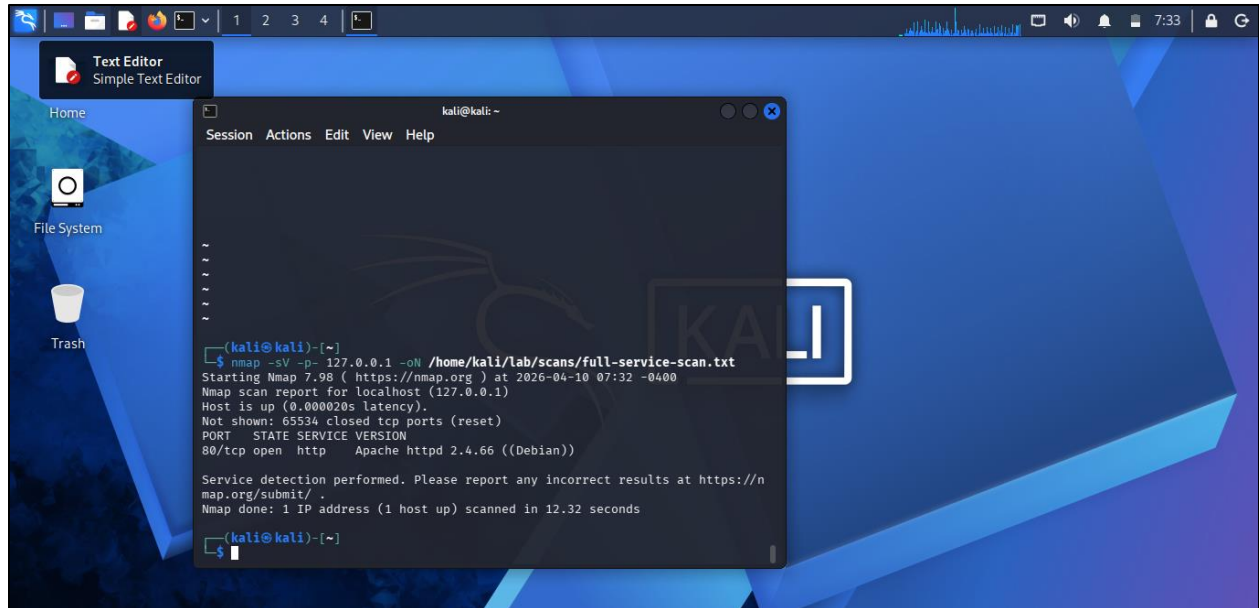
Nmap scan was performed again:

```
nmap -sV -p- 127.0.0.1 -oN /home/kali/lab/scans/full-service-scan.txt
```

Observation:

- Port **80/tcp** was found open
- Service: Apache HTTP Server (2.4.66)





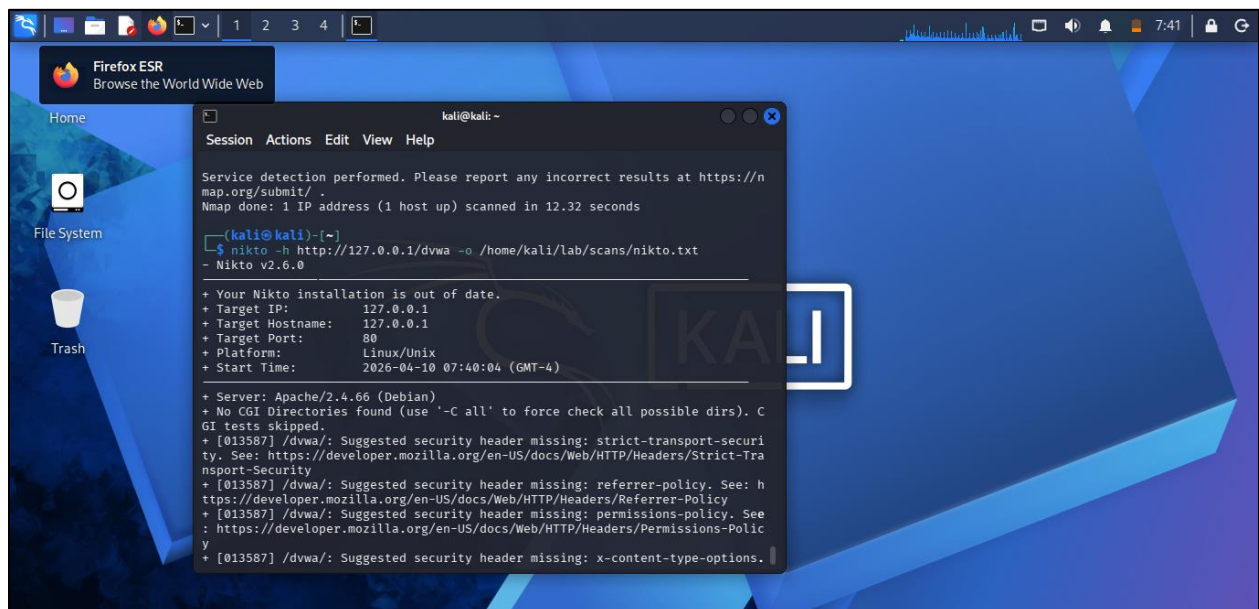
Step 8: Web Vulnerability Scanning (Nikto)

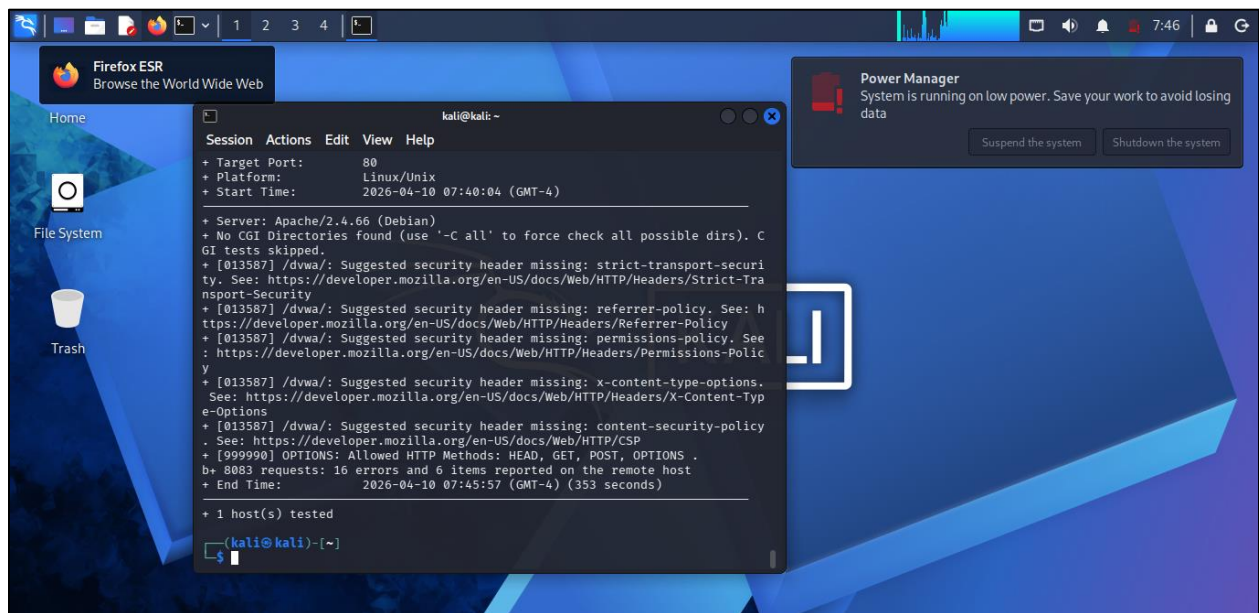
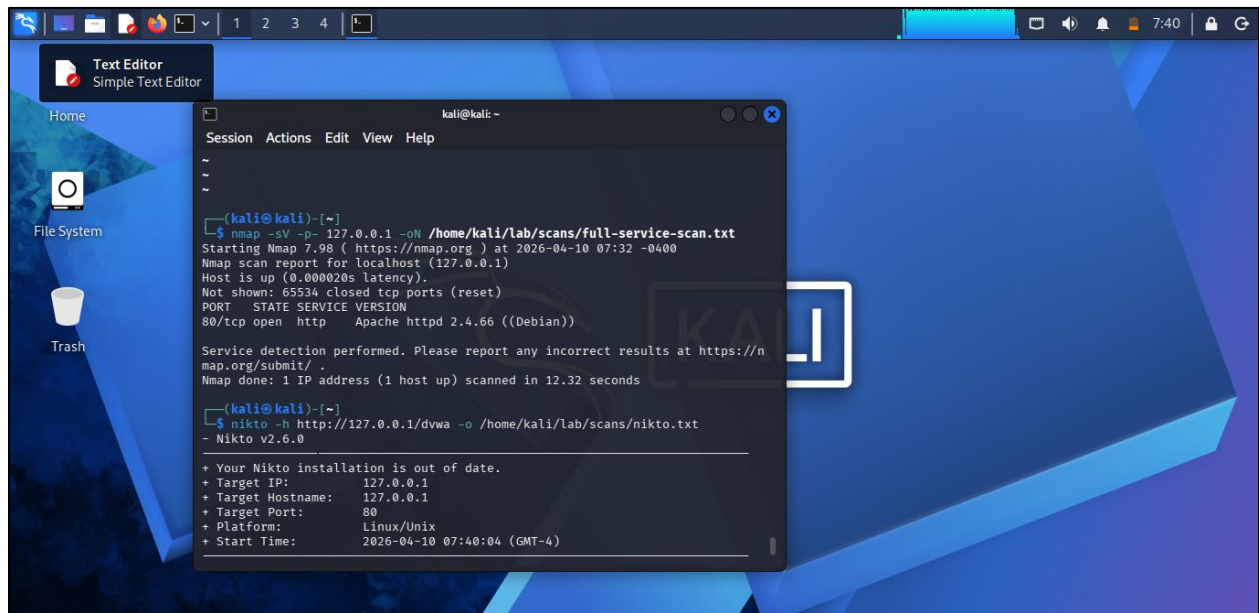
Nikto scan was performed:

```
nikto -h http://127.0.0.1/dvwa -o /home/kali/lab/scans/nikto.txt
```

Observation:

- Missing security headers detected
- HTTP methods identified

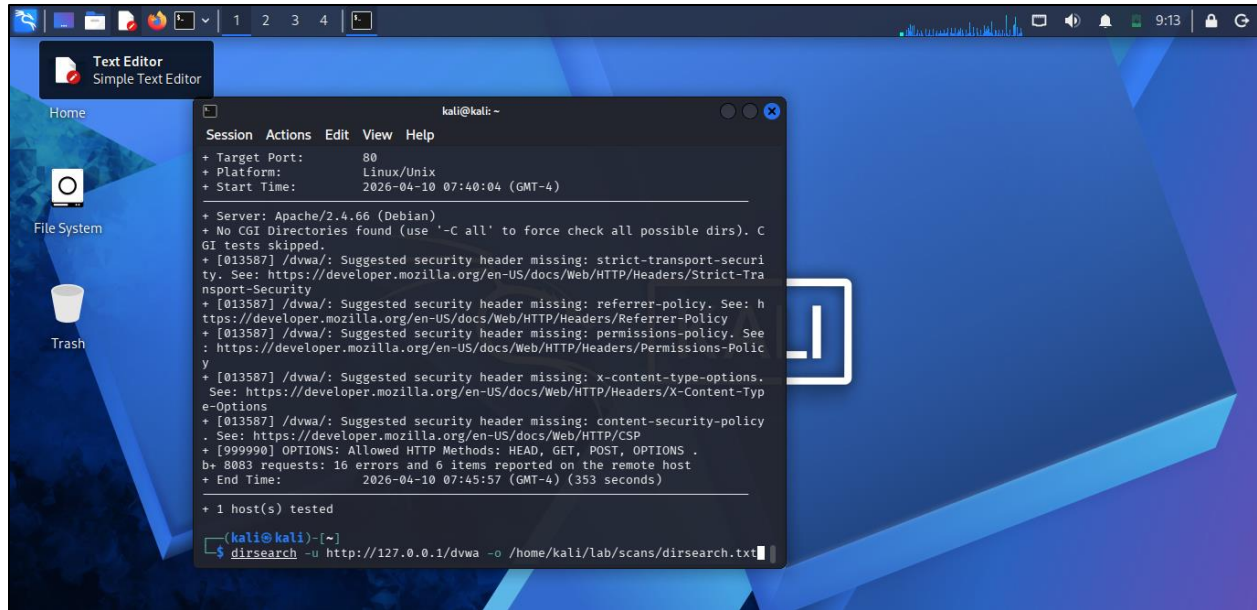




Step 9: Directory Enumeration (Dirsearch Installation)

Dirsearch tool was not available initially, so it was installed:

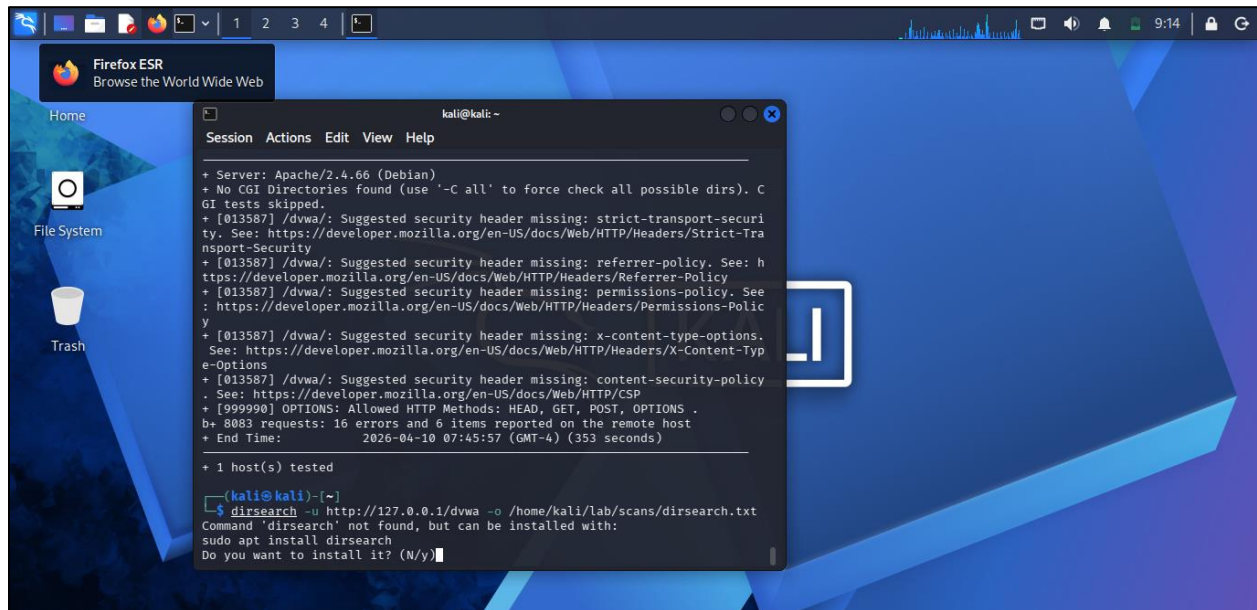
```
sudo apt install dirsearch
```

```
kali@kali:~$ dirsearch -u http://127.0.0.1/dvwa -o /home/kali/lab/scans/dirsearch.txt
```

Session Actions Edit View Help

- + Target Port: 80
- + Platform: Linux/Unix
- + Start Time: 2026-04-10 07:40:04 (GMT-4)
- + Server: Apache/2.4.66 (Debian)
- + No CGI Directories found (use '-C all' to force check all possible dirs). CGI tests skipped.
- + [013587] /dvwa/: Suggested security header missing: strict-transport-security. See: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-Transport-Security>
- + [013587] /dvwa/: Suggested security header missing: referrer-policy. See: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Referrer-Policy>
- + [013587] /dvwa/: Suggested security header missing: permissions-policy. See: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Policy>
- + [013587] /dvwa/: Suggested security header missing: x-content-type-options. See: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Content-Type-Options>
- + [013587] /dvwa/: Suggested security header missing: content-security-policy. See: <https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP>
- + [999990] OPTIONS: Allowed HTTP Methods: HEAD, GET, POST, OPTIONS . b+ 8083 requests: 16 errors and 6 items reported on the remote host
- + End Time: 2026-04-10 07:45:57 (GMT-4) (353 seconds)
- + 1 host(s) tested



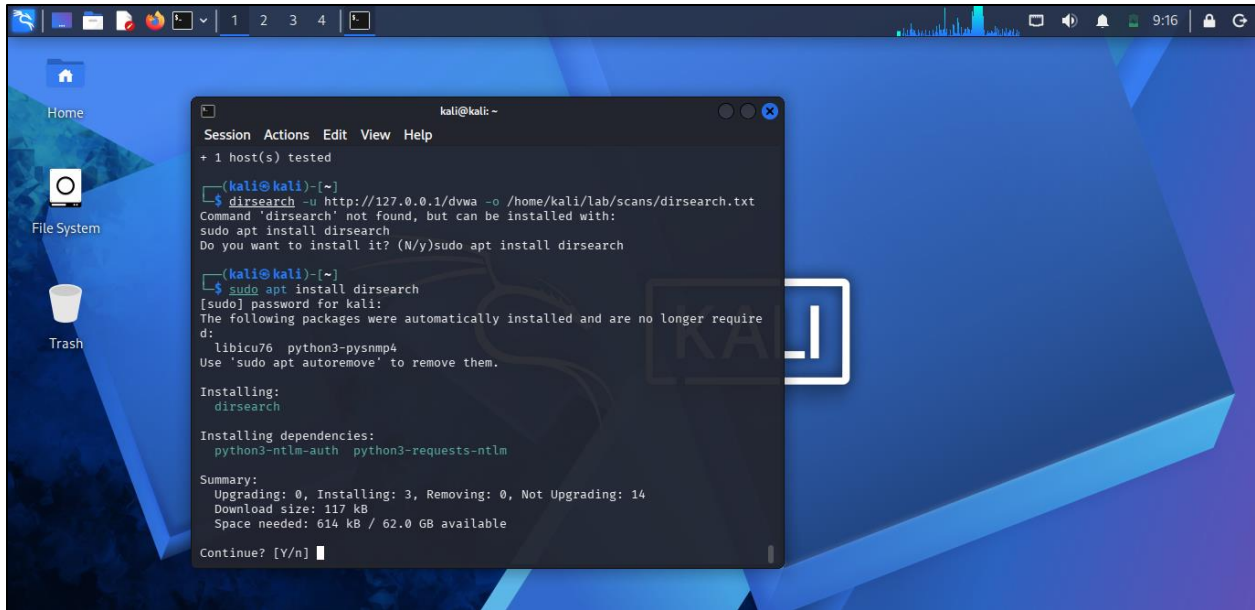
```
kali@kali:~$ dirsearch -u http://127.0.0.1/dvwa -o /home/kali/lab/scans/dirsearch.txt
```

Session Actions Edit View Help

- + Server: Apache/2.4.66 (Debian)
- + No CGI Directories found (use '-C all' to force check all possible dirs). CGI tests skipped.
- + [013587] /dvwa/: Suggested security header missing: strict-transport-security. See: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-Transport-Security>
- + [013587] /dvwa/: Suggested security header missing: referrer-policy. See: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Referrer-Policy>
- + [013587] /dvwa/: Suggested security header missing: permissions-policy. See: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Policy>
- + [013587] /dvwa/: Suggested security header missing: x-content-type-options. See: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Content-Type-Options>
- + [013587] /dvwa/: Suggested security header missing: content-security-policy. See: <https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP>
- + [999990] OPTIONS: Allowed HTTP Methods: HEAD, GET, POST, OPTIONS . b+ 8083 requests: 16 errors and 6 items reported on the remote host
- + End Time: 2026-04-10 07:45:57 (GMT-4) (353 seconds)
- + 1 host(s) tested

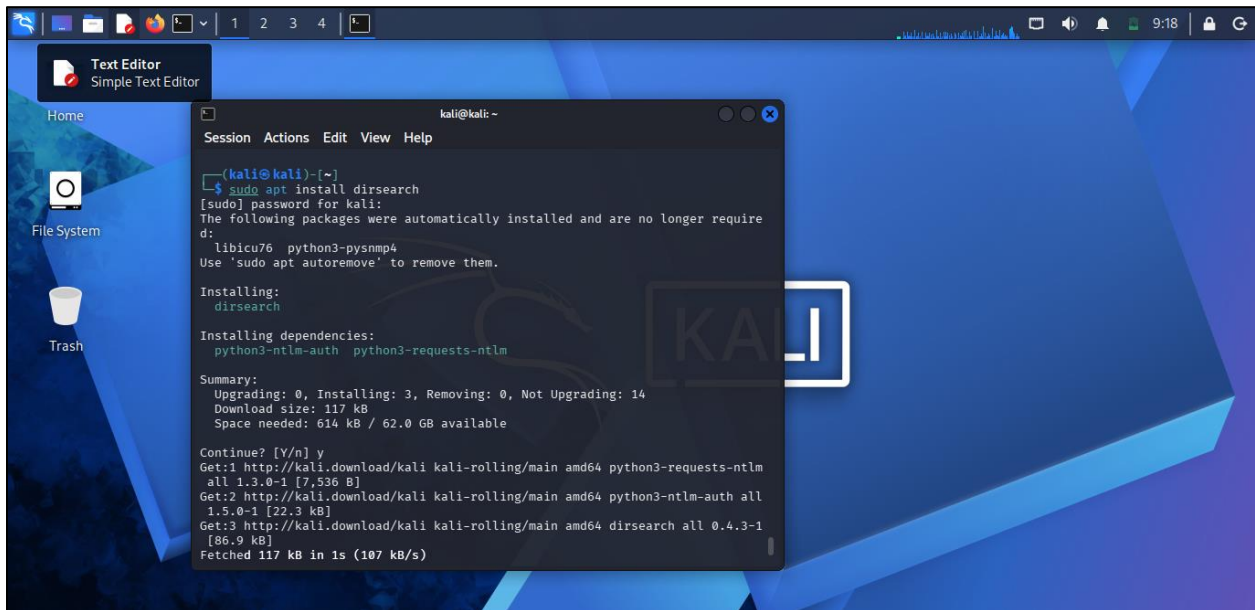
(kali@kali)~\$ dirsearch -u http://127.0.0.1/dvwa -o /home/kali/lab/scans/dirsearch.txt

Command 'dirsearch' not found, but can be installed with:
sudo apt install dirsearch
Do you want to install it? (N/y)



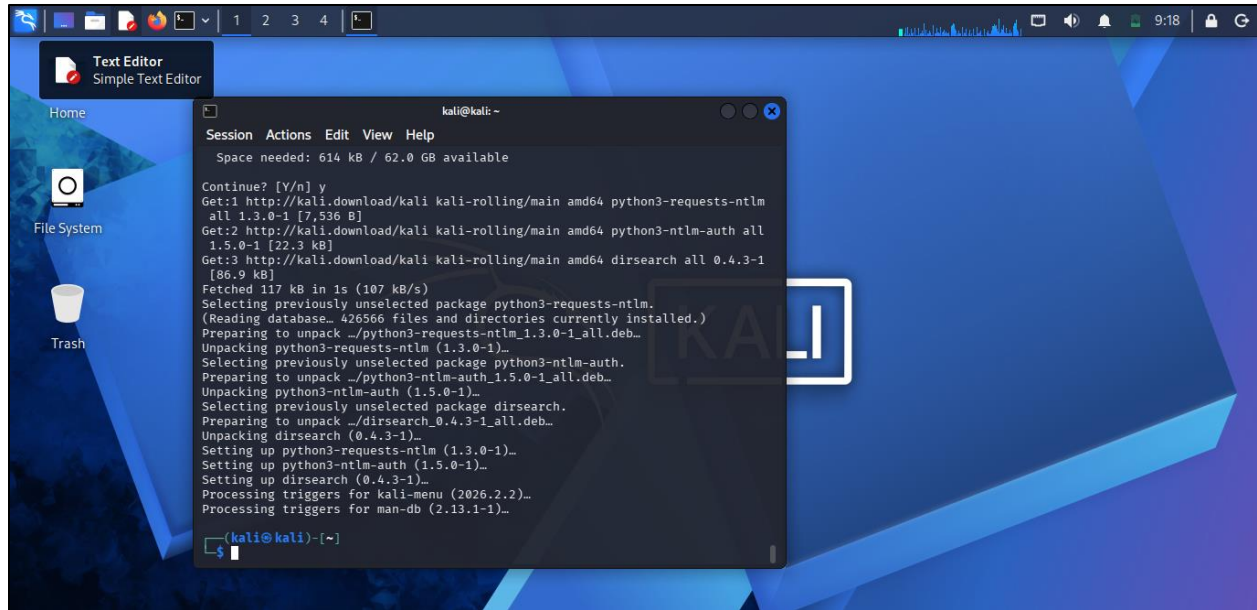
The terminal window shows the user attempting to run the `dirsearch` command. It reports that the command is not found and suggests installing it via `sudo apt install dirsearch`. The user confirms the installation, and the system lists the packages to be installed along with their dependencies.

```
kali@kali: ~  
$ dirsearch -u http://127.0.0.1/dvwa -o /home/kali/lab/scans/dirsearch.txt  
Command 'dirsearch' not found, but can be installed with:  
sudo apt install dirsearch  
Do you want to install it? (N/y) sudo apt install dirsearch  
  
[kali@kali]~$ sudo apt install dirsearch  
[sudo] password for kali:  
The following packages were automatically installed and are no longer required:  
libicu76 python3-pysnmp4  
Use 'sudo apt autoremove' to remove them.  
  
Installing:  
dirsearch  
  
Installing dependencies:  
python3-ntlm-auth python3-requests-ntlm  
  
Summary:  
Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 14  
Download size: 117 kB  
Space needed: 614 kB / 62.0 GB available  
  
Continue? [Y/n]
```



The terminal window shows the completion of the installation process. The user presses 'y' to confirm, and the system proceeds to download and install the required packages. The output shows the progress of the installation, including the download of `python3-requests-ntlm`, `python3-ntlm-auth`, and `dirsearch`.

```
kali@kali: ~  
$ sudo apt install dirsearch  
[sudo] password for kali:  
The following packages were automatically installed and are no longer required:  
libicu76 python3-pysnmp4  
Use 'sudo apt autoremove' to remove them.  
  
Installing:  
dirsearch  
  
Installing dependencies:  
python3-ntlm-auth python3-requests-ntlm  
  
Summary:  
Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 14  
Download size: 117 kB  
Space needed: 614 kB / 62.0 GB available  
  
Continue? [Y/n] y  
Get:1 http://kali.download/kali kali-rolling/main amd64 python3-requests-ntlm  
all 1.3.0-1 [7,536 B]  
Get:2 http://kali.download/kali kali-rolling/main amd64 python3-ntlm-auth all  
1.5.0-1 [22.3 kB]  
Get:3 http://kali.download/kali kali-rolling/main amd64 dirsearch all 0.4.3-1  
[86.9 kB]  
Fetched 117 kB in 1s (107 kB/s)
```

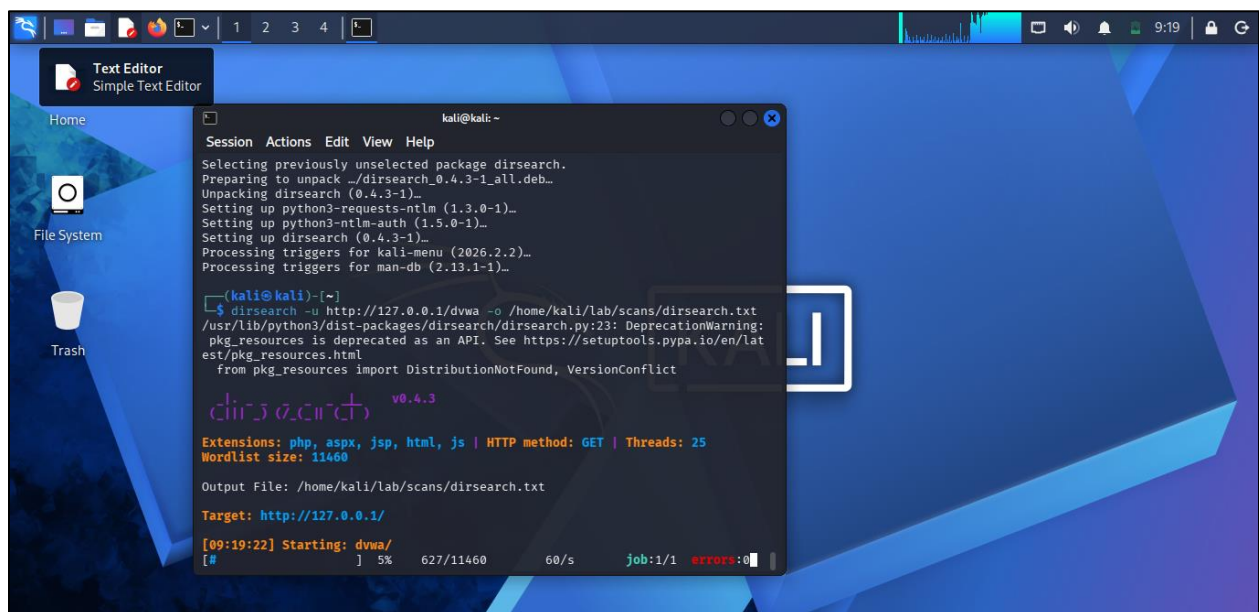
Step 10: Directory Enumeration (Dirsearch Scan)

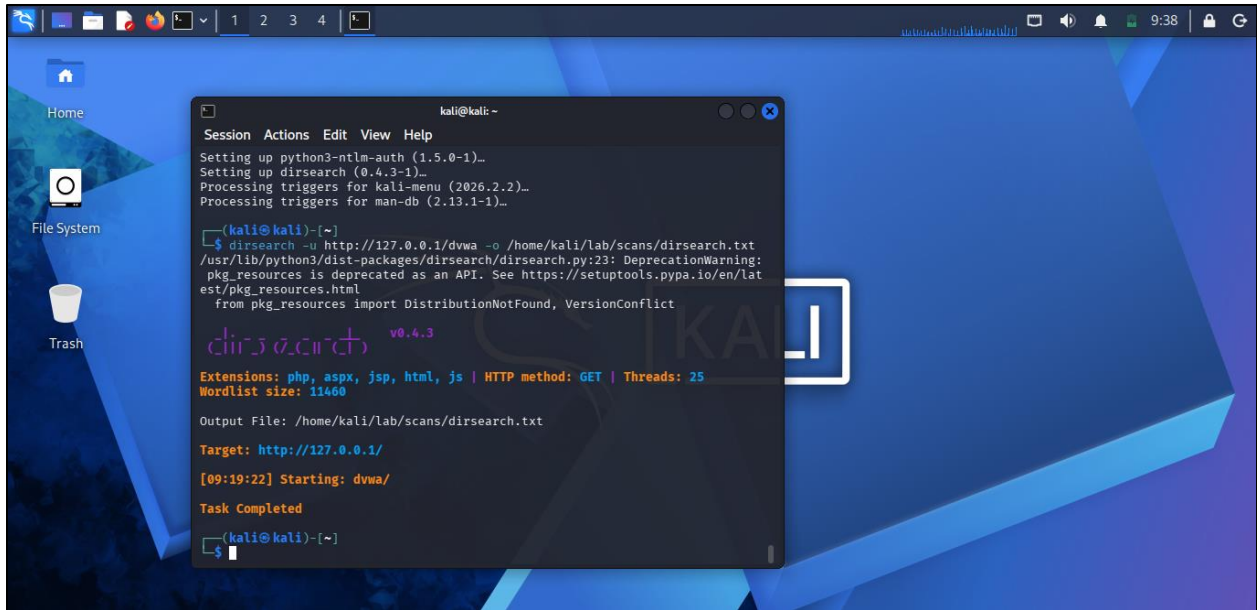
Dirsearch scan was executed:

```
dirsearch -u http://127.0.0.1/dvwa -o /home/kali/lab/scans/dirsearch.txt
```

Observation:

- No additional directories were discovered





4. Technical Findings

Finding 1 — Open HTTP Port

- **Observation:**
Port 80/tcp is open running Apache HTTP Server (2.4.66)
- **Risk Level:** Medium
- **Recommendation:**
Secure and restrict access to web services

Finding 2 — Missing Security Headers

- **Observation:**
Missing headers:
 - HSTS
 - Referrer Policy
 - Permissions Policy
 - X-Content-Type-Options
 - CSP
- **Risk Level:** Medium
- **Recommendation:**
Implement proper HTTP security headers

Finding 3 — Insecure HTTP Methods Enabled

- **Observation:**
Allowed methods: GET, POST, HEAD, OPTIONS
- **Risk Level:** Low–Medium
- **Recommendation:**
Restrict unnecessary HTTP methods

Finding 4 — Directory Enumeration Result

- **Observation:**
No directories found during scan
- **Risk Level:** Informational

Finding 5 — Vulnerable Application (DVWA)

- **Observation:**
DVWA is intentionally insecure
- **Risk Level:** High

5. Impact Analysis

The system exposes a web server with certain misconfigurations. Missing headers and open services increase the attack surface and may allow attackers to identify vulnerabilities.

6. Risk Rating Summary

Finding	Risk Level
Open HTTP Port	Medium
Missing Security Headers	Medium
Insecure HTTP Methods	Low–Medium
Directory Enumeration	Informational
DVWA Application	High

7. Remediation

- Secure web server configuration
- Implement security headers
- Restrict HTTP methods
- Regular vulnerability scans
- Avoid using vulnerable apps in production

8. Conclusion

The vulnerability assessment was successfully performed in a controlled lab environment. Tools like Nmap, Nikto, and Dirsearch helped identify key security weaknesses. This practical provided hands-on experience in vulnerability scanning and cybersecurity analysis.